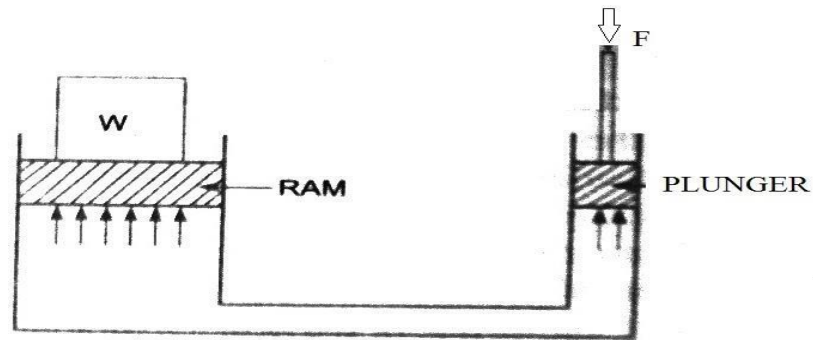


Fluid system

Hydraulic press

- The hydraulic press is a device used for lifting heavy weights by the application of a much smaller force.
- It is based on Pascal's law, which states that the intensity of pressure in a static fluid is transmitted equally in all direction



Construction

- It consists of two cylinders of different diameter.
- One of the cylinders is of large diameter and contains a ram.
- The other cylinder is of smaller diameter and contains a plunger.
- The two cylinders connected by pipe.
- The cylinder and pipe contain a liquid through which pressure is transmitted.

Working

- When a small force F is applied on the plunger in the downward direction,
- A pressure is produced on the liquid in contact with the plunger.
- This pressure is equally transmitted equally in all direction and acts on the ram in the upward direction.
- The heavier weight placed on the ram is then lifted up.

Let

W = Weight to be lifted

F = force applied on the plunger

A = Area of ram

a = area of plunger

p = pressure intensity produced by force F

$$= \text{—————} = -$$

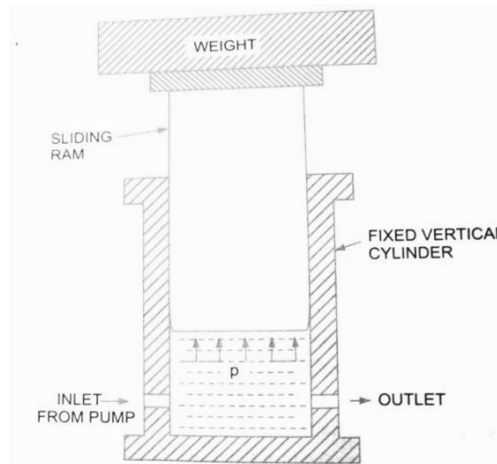
Pressure intensity of ram (p) = —

Equating the pressure intensity, = —

$$W = \pi A$$

Hydraulic accumulator

- The hydraulic accumulator is device used for storing the energy of the liquid in the form of pressure energy, which may be supplied for any sudden or intermittent requirement.
- In case of hydraulic lift or hydraulic crane, a large amount of energy is required when lift or crane is moving upward.
- This energy is supplied from accumulator.



Construction

- It consists of fixed vertical cylinder containing a sliding ram.
- A heavy weight is placed on the ram.
- The inlet of the cylinder connected to the pump.
- The outlet of the cylinder is connected to the machine

Working

- The ram is at the lowermost position in the beginning.
- The pump supplies water under pressure continuously.
- If the water under pressure is not required by the machine, the water under pressure will be stored in the cylinder.
- This will raise the ram on which a heavy weight is placed.
- When the ram is at uppermost position, the cylinder is full of water and accumulator is stored the maximum amount of pressure energy.
- When they require a large amount of energy, the hydraulic accumulator will supply this energy and ram will move in downward direction.

Capacity of the accumulator

- It is defined as the maximum amount of hydraulic energy stored in the accumulator.

$$\text{Weight} = P \times A$$

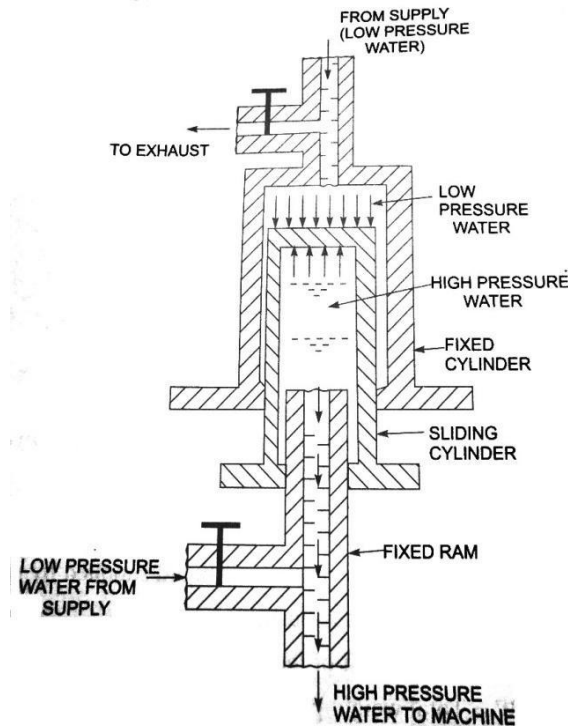
$$\text{Capacity} = P \times A \times L$$

Where, p = intensity pressure, A = area of the sliding ram, L = stroke or lift of the ram.

$$\text{Volume} = A \times L$$

Hydraulic intensifier

- The device which is used to increase the intensity of pressure of water by means of hydraulic energy available from large amount of water at a low pressure is called hydraulic intensifier.
- Such a device is needed when the hydraulic machines such as hydraulic press requires water at very high pressure which cannot be obtained from the main supply directly.



Construction:

- It consists of fixed ram through which the water, under a high pressure, flows to the machine.
- A hollow inverted sliding cylinder, containing water under high pressure, is mounted over the fixed ram.
- The inverted sliding cylinder is surrounded by another fixed inverted cylinder which contains water from the main supply at a low pressure.

Working

- A large quantity of water at low pressure from supply enters the inverted fixed cylinder.
- The weight of this water pressures the sliding cylinder in the downward direction.
- The water in the sliding cylinder gets compressed due to the downward movement of the sliding cylinder and its pressure is thus increased.
- The high pressure water is forced out of the sliding cylinder through the fixed ram, to the machine.

Let,

p = intensity of pressure of water supply to the fixed cylinder

A = external area of the sliding cylinder

a = area of the end of the fixed ram

P^* = intensity of pressure of water in the sliding cylinder

Force exerted by low pressure water = $p \times A$

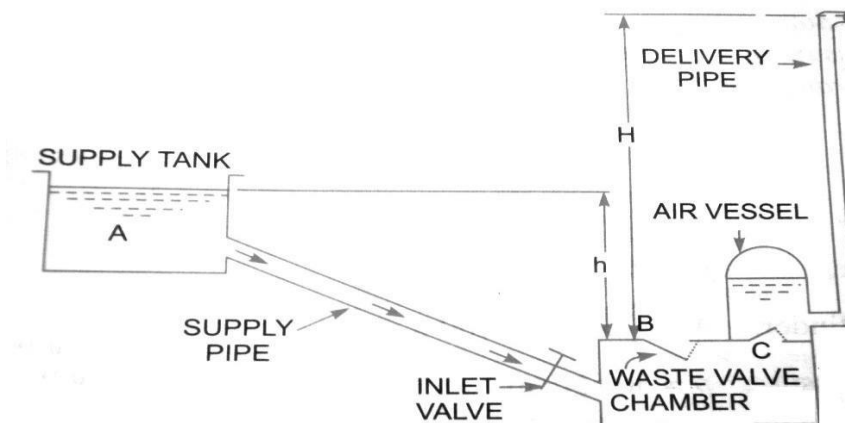
Force exerted by high pressure water = $P^* \times a$

Equating the force,

$$p \times A = P^* \times a$$

Hydraulic Ram

- The hydraulic ram is a pump which raises water without any external power for its operation.
- When large quantity of water is available at small height, a small quantity of water is raised to a greater height with the help of hydraulic ram.
- It works under the principle of hydraulic hammer.



Construction

- It consists of supply tank which large quantity of water is available at small height.
- A supply pipe connected in between the supply tank and the chamber
- Inlet valve is at the end of the supply pipe
- Delivery pipe is connected to the chamber having a small quantity of water at a greater height.

Working

- When the inlet valve fitted to the supply pipe is opened, water flows from supply tank to chamber, which has two valves B and C.
- The valve B is called waste valve and valve C is called delivery valve. The valve C is fitted to the air vessel.
- As the water is coming into the chamber from supply tank, the level of water raises in the chamber and waste valve B moving upward.
- A stage comes, when the waste valve B suddenly closes.
- This sudden closure of waste valve creates high pressure inside the chamber. This high pressure opens the delivery valve C.
- The water from the chamber enters the air vessel and compresses the air inside the air vessel.
- This compressed air exerts force on the air vessel and small quantity of water is raised to greater height.
- When the water in the chamber loses its momentum, the waste valve B opens in downward direction and the flow of water from supply tank starts flowing to the chamber and cycle will be repeated.

W = weight of water is flowing per second into the chamber

w = weight of water raised per second

h = height of water in the supply tank above the chamber

H = height of water raised from the chamber.

Energy supplied by the supply tank to ram = $W \times h$

Energy delivered by the ram = $w \times H$

D' Aubuisson's Efficiency = $\frac{w \times H}{W \times h}$

Rankine Efficiency = $\frac{w \times H}{W \times h}$

Where q = discharge of delivery pipe

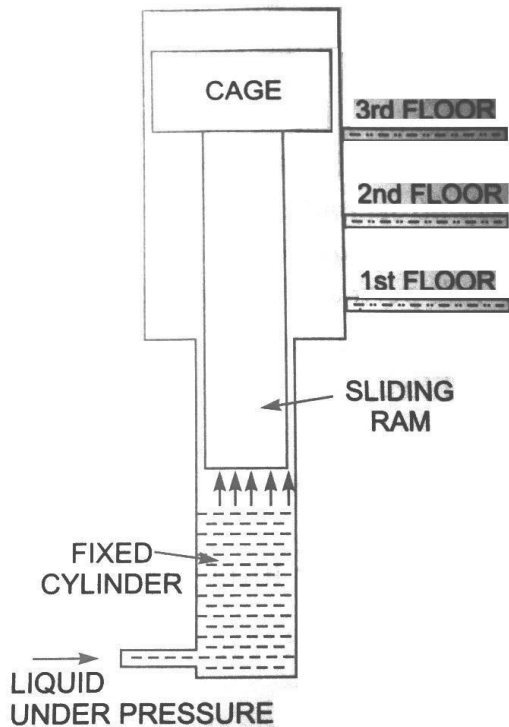
Q = discharge through supply pipe

Hydraulic Lift

The hydraulic lift is a device used for carrying passenger or goods from one floor to another in multi-storeyed building. The hydraulic lifts are two types, namely

- a. Direct acting hydraulic lift
- b. Suspended hydraulic lift

a. Direct acting hydraulic lift



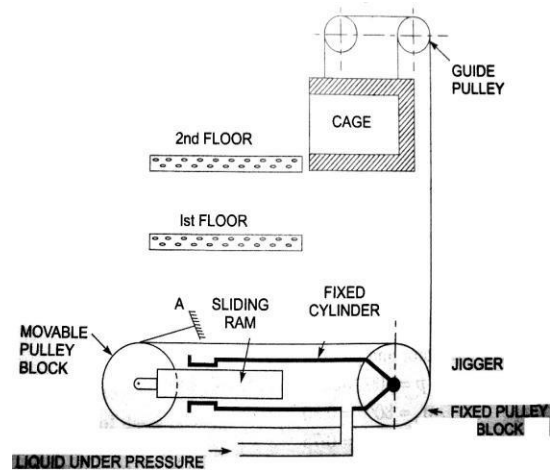
Construction:

- It consists of a ram, sliding in fixed cylinder.
- At the top of the sliding ram, cage is fitted.

Working:

- The liquid under pressure flows into the fixed cylinder.
- This liquid exerts force on the sliding ram, which moves vertically up and thus raises the cage to the required height.
 - The cage is moving down by removing the liquid from the fixed cylinder

b. Suspended hydraulic lift



Construction:

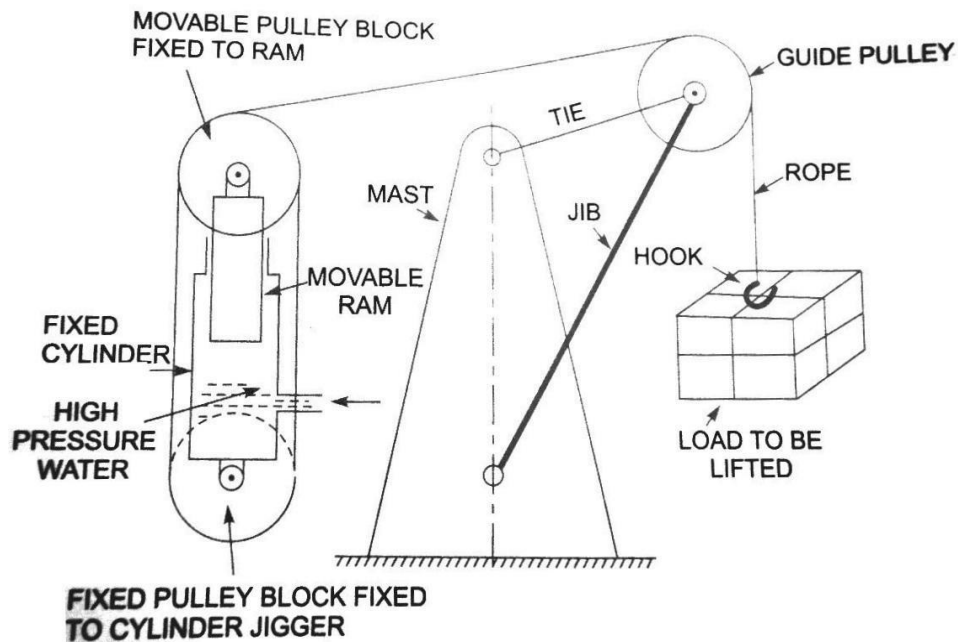
- It consists of cage, which is suspended from a wire rope.
- A jigger consisting of fixed cylinder, a sliding ram and a set of two pulley blocks is provided at the foot
- One of the pulley blocks is movable and other is fixed.
- Movable pulley is connected at the end of the sliding ram.

Working:

- When water under high pressure is admitted into the fixed cylinder of the jigger, the sliding ram is forced to move towards left.
- Thus increase the distance between two pulley blocks.
- The wire rope connected to the cage is pulled and cage is lifted.
- When the water is taken away from the cylinder the cage is down, due decrease the distance from the two pulleys.

Hydraulic crane:

- Hydraulic crane is a device, used for raising or transferring heavy loads.
- It is widely used in workshop, ware house and dock sidings.



Construction:

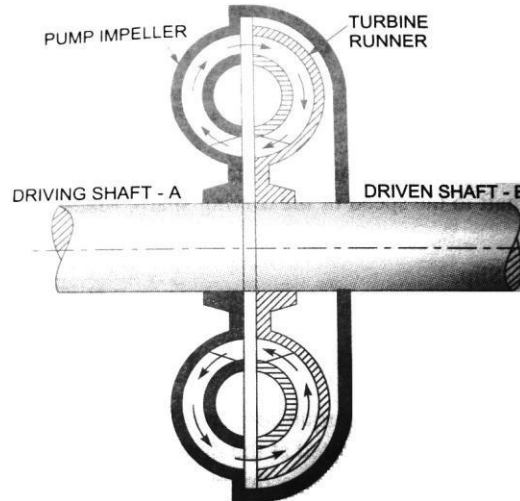
- It consists of mast, tie, jib, guide pulley and a jigger.
- The jib and tie are attached to the mast.
- The jigger which consists of a movable ram sliding in the fixed cylinder is used for lifting or lowering the heavy loads.
- One end of the ram is contact with water and the other end is connected to a set of movable pulley block.
- Another pulley block is called the fixed pulley block is attached to the fixed cylinder, is not having any movement
- The wire rope, one end is connected to movable pulley, other is connected to hook through fixed pulley and guide pulley.

Working:

- The water under high pressure is admitted into the cylinder of the jigger.
- This water forces the sliding ram to move vertically up.
- Due to the movement of the ram in the vertical up direction, the movable pulley block attached to the ram also moves upward.
- This increases the distance between two pulley blocks and hence the wire passing over the guide pulley is pulled by the jigger. This raises the load attached to the hook.

Fluid or Hydraulic coupling

- It is device used for transmitting power from driving shaft to driven shaft with the help of fluid.
- There is no mechanical connection between two shafts.



Construction:

- It consists of radial pump impeller mounted on the driving shaft A and radial flow reaction turbine mounted on the driven shaft B.
- Both the impeller and runner are in identical shape and they together form a casing which is completely enclosed and filled with oil.

Working:

- In the beginning, both the shafts A and B are at rest. When the driving shaft A is rotated, the oil starts moving from the inner radius to the outer radius of the pump impeller.
- The pressure energy and kinetic energy of the oil get increases at the outer radius of the pump impeller.
- This oil of increased energy enters the runner of the reaction turbine at the outer radius of the turbine runner and flows inwardly to the inner radius of the turbine runner.
- The oil, while flowing through the runner, transfers its energy to the blades of the runner and makes the runner to rotate.
- The oil, from the runner flows back to the pump impeller, thus having a continuous circulation.

Efficiency of the fluid coupling = _____

= _____

Power at any shaft = _____

Let N_A = speed of shaft A,

N_B = speed of shaft B,

T_A = Torque at shaft A,

T_B = Torque at shaft B.

Efficiency = $(N_B \times T_B) / (N_A \times T_A)$

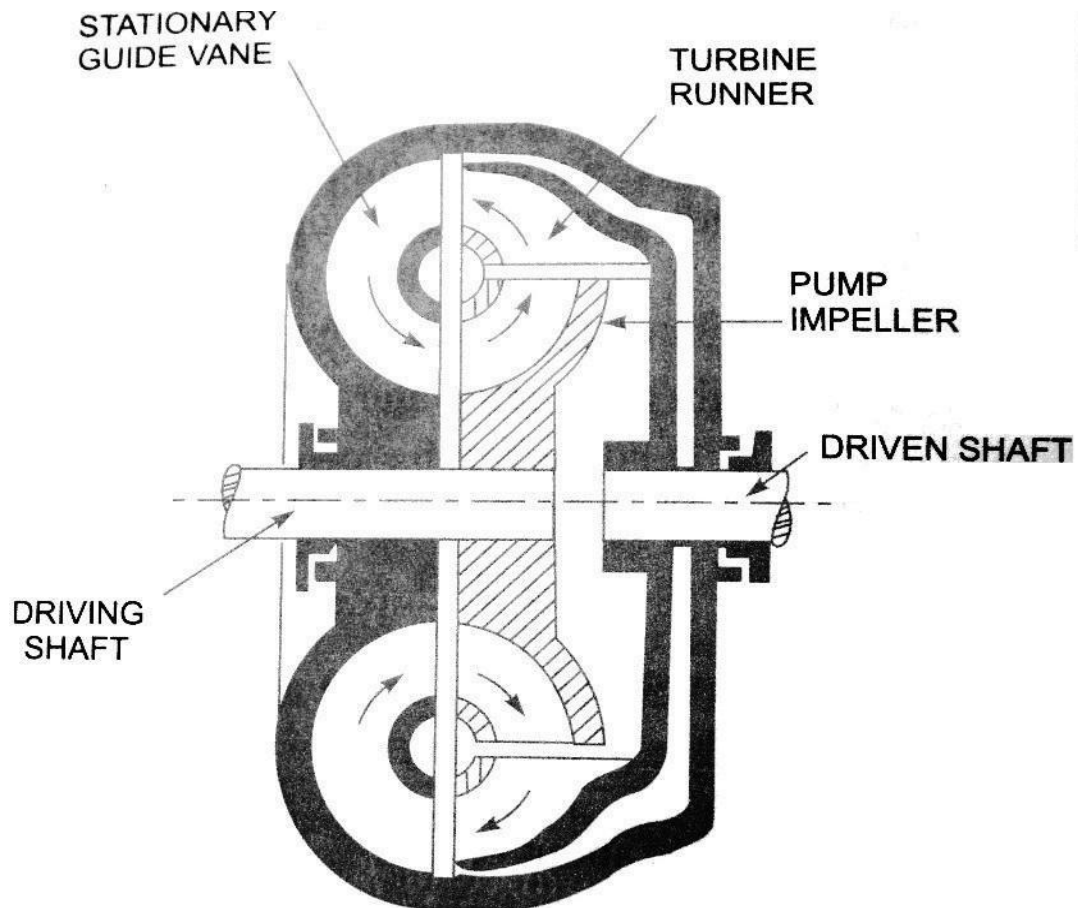
$T_A = T_B$ (torque transmitted is same)

Efficiency = N_B / N_A

Slip = $(N_A - N_B) / N_A$

Hydraulic Torque Converter

- It is a device used for transmitting increased torque at the driven shaft.
- The torque transmitted at the driven shaft may be more or less than the torque available at the driving shaft.
- The torque at the driven shaft may be increased about five times the torque available at the driving shaft with an efficiency of 90 %.



Construction:

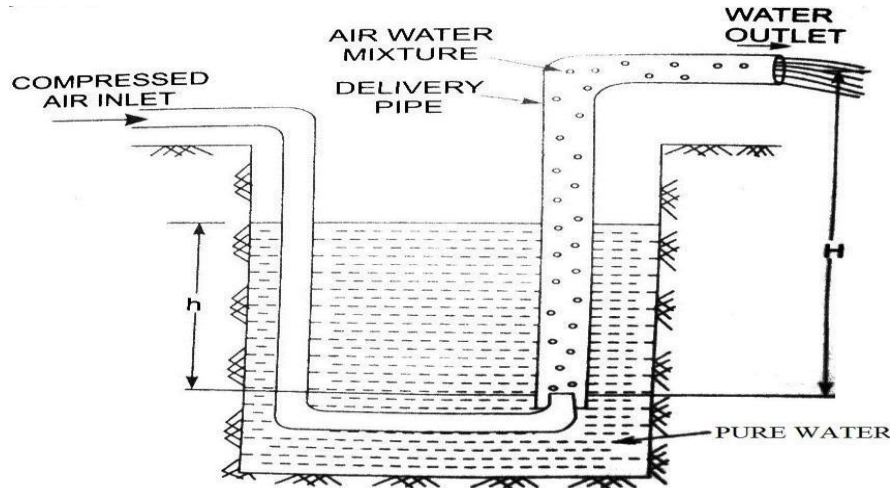
- It consists of driving shaft and driven shaft.
- Stationary guide vanes, pump impeller and turbine impeller

Working

- The power at any shaft is proportional to the product of the torque and the speed of the shaft.
- Hence if torque at the driven shaft is to be increased, the corresponding value of the speed at the same shaft should be decreased.
- The speed of the driven shaft is decreased by decreasing the velocity of the oil, which is allowed to flow from the pump impeller to the turbine runner and then through stationary guide vane.
- Due to decrease in speed at the driven shaft, the torque increases

The Air Lift Pump:

- The air lift pump is a device which is used for lifting water from well or sump by using compressed air.
- The compressed air is made to mix with water. The density of the mixture of air and water is reduced.
- The density of the mixture is much less than that of pure water. Hence a very small column of pure water can balance a very long column of air water mixture.
- Thus is the principle on which the airlift pump works.



Construction:

- It consists of delivery pipe having air water mixture.
- Pure water was in a fixed tank, compressed air at inlet pipe and water at outlet pipe.

Working:

- The compressed air is introduced through one or more nozzles at the foot of the delivery pipe, which is fixed in the well from which water is to be lifted.
- In the delivery pipe, a mixture of air and water is formed.
- The density of air water mixtures becomes very less as compared to the density of pure water.
- Hence a small column of pure water will balance a very long column of air water mixture.
- This air water mixture will be discharged out of delivery pipe. The flow will continue as long as there is supply of compressed air.

Advantage:

- The air lift pump is not having any moving parts below water level and hence there are no chances of suspended solid particles damaging the pump.
- This pump can raise more water through a bore hole of given diameter than any other pump

Disadvantage:

- The efficiency of the pump is low only 20 to 40 %.
- The discharge is minimum compared to other pump.

FLUID SYSTEM

1. A hydraulic press has a ram of 300 mm diameter and a plunger of 45 mm diameter. Find the weight lifted by the hydraulic press when the force applied at the plunger is 50 N. If the stroke of the plunger is 100 mm, find the distance travelled by the weight in 100 strokes. Determine the work done during 100 strokes.

Given:

$$D = 0.3 \text{ m}$$

$$d = 0.045 \text{ m}$$

$$F = 50 \text{ N}$$

$$\text{Stroke of the plunger}(X) = 0.1 \text{ m}$$

$$\text{No of strokes} = 100$$

To Find:

$$W = ?$$

$$W.D = ?$$

Solution:

$$A = \frac{\pi D^2}{4} = 0.07068 \text{ m}^2$$

$$a = \frac{\pi d^2}{4} = 0.00159 \text{ m}^2$$

$$W = \frac{F}{a} A = 2222.64 \text{ N}$$

$$\text{Let distance moved by the ram} = Y$$

$$A \times Y = a \times X$$

$$Y = 0.00225 \text{ m}$$

$$\text{Distance moved in 100 strokes} = 100 \times Y = 0.225 \text{ m}$$

$$W.D = \text{Weight} \times \text{Distance moved} = 500 \text{ Nm}$$

2. A hydraulic press has a ram of 200 mm diameter and a plunger of 30 mm diameter. It is used to lifting a weight of 3KN. Find the force required at the plunger. If a lever is used for applying force on the plunger, find the force required at the end of the lever if the ratio l/L is 1/10.

Given:

$$D = 0.2 \text{ m}$$

$$d = 0.03 \text{ m}$$

$$W = 3000 \text{ N}$$

$$- = -$$

To Find:

$$F = ?$$

$$F' = ?$$

Solution:

$$A = \frac{\pi D^2}{4} = 0.0314 \text{ m}^2$$

$$a = \frac{\pi d^2}{4} = 7.068 \times 10^{-4} \text{ m}^2$$

$$F = \frac{W}{A} a = 67.52 \text{ N}$$

$$F' = \frac{F}{l/L} = 675.2 \text{ N}$$

3. A hydraulic press has a ram of 150 mm diameter and a plunger of 20 mm diameter. The stroke of the plunger is 200 mm and weight lifted is 800 N. If the distance moved by the weight is 1m in 20 min determines: (a) The force applied on the plunger, (b) Power required to drive the plunger, (c) Number of strokes performed by the plunger.

Given:

$$D = 0.15 \text{ m}$$

$$d = 0.02 \text{ m}$$

$$W = 800 \text{ N}$$

$$\text{Stroke of the plunger}(X) = 0.2 \text{ m}$$

$$\text{Distance moved by the weight}(Y) = 1 \text{ m}$$

$$\text{Time} = 20 \text{ min} = 120 \text{ seconds}$$

To Find:

$$F = ?$$

$$P = ?$$

$$\text{No of strokes performed by the plunger} = ?$$

Solution:

$$A = \frac{\pi D^2}{4} = 0.01767 \text{ m}^2$$

$$a = \frac{\pi d^2}{4} = 3.1416 \times 10^{-4} \text{ m}^2$$

☐ **Force applied on the plunger**

$$F = \frac{W}{A} a = 14.22 \text{ N}$$

☐ **Power required to drive the plunger:**

$$W.D = \frac{W \times Y}{\text{Time}} = 0.6667 \text{ Nm/s}$$

$$\text{Power} = \frac{W.D}{\text{Time}} = 0.000667 \text{ kW}$$

☐ **No of strokes performed by the plunger:**

$$\text{No of strokes} = \frac{Y}{\text{Stroke}} = 5$$

4. Determine the length of stroke for an accumulator having a displacement of 115 litres. The diameter of the plunger is 350 mm.

Given:

$$V = 0.115 \text{ m}^3$$

$$D = 0.350 \text{ m}$$

To Find:

$$L = ?$$

Solution:

$$A = \frac{\pi D^2}{4} = 0.09621 \text{ m}^2$$

$$V = A \times L$$

$$L = 1.195 \text{ m}$$

5. The water is supplied at a pressure of 14 N/cm^2 to an accumulator, having a ram of diameter 1.5 m. If the total lift of the ram is 8 m, determine: (a) The capacity of the accumulator, (b) Total weight placed on the ram.

Given:

$$p = 14 \times 10^4 \text{ N/m}^2$$

$$D = 1.5 \text{ m}$$

$$L = 8 \text{ m}$$

To find:

Capacity =?

Total weight =?

Solution:

□ **Capacity of the accumulator:**

$$= p \times A \times L = 1979 \text{ kNm}$$

□ **Total weight placed on the ram:**

$$W = p \times L = 247380 \text{ N}$$

6. Total weight (including the self weight of ram) placed on the sliding ram of a hydraulic accumulator is 40 kN. The diameter of the ram is 500 mm. If the frictional resistance against the movement of the ram is 5% of the total weight, determine the intensity of pressure of water when: (a) The ram is moving up with a uniform velocity, (b) The ram is moving down with uniform velocity.

Given:

$$W = 40000 \text{ N}$$

$$D = 0.5 \text{ m}$$

$$\text{Frictional resistance} = 0.05$$

$$W_{\text{act}} = p \times A$$

$$A = \frac{\pi}{4} D^2 = 0.1963 \text{ m}^2$$

$$P = 21.4 \times 10^4 \text{ N/m}^2$$

To find:

Intensity pressure- when ram moving up =?

Intensity pressure- when ram moving down =?

Solution:

□ **Intensity pressure when ram moving up:**

Actual weight when ram moving up

$$= W \times 1.05 = 42000 \text{ N}$$

□ **Intensity pressure when ram moving down:**

Actual weight when ram moving down

$$= W \times 0.95 = 38000 \text{ N}$$

$$W_{\text{act}} = p \times A$$

$$A = \frac{\pi}{4} D^2 = 0.1963 \text{ m}^2$$

$$P = 19.35 \times 10^4 \text{ N/m}^2$$

7. An accumulator is loaded with 40 kN weight. The ram has a diameter of 30 cm and stroke of 6 m. Its friction may be taken as 5%. It takes 2 min to **fall through** its full stroke. Find the total work supplied and power delivered to the hydraulic appliance by the accumulator, when 7.5 lit/s is being delivered by a pump, while the accumulator descends with the stated velocity.

Given:

$$W = 40000 \text{ N}$$

$$D = 0.3 \text{ m}$$

$$\text{Distance moved (L)} = 6 \text{ m}$$

$$Q = 0.0075 \text{ m}^3/\text{s}$$

$$\text{Frictional resistance} = 0.05$$

$$\text{Time} = 120 \text{ sec}$$

Work of accumulator

$$\text{Actual weight} = W \times 0.95 = 38000 \text{ N}$$

$$W.D_{\text{acc}} = \text{—————} = 1900 \text{ Nm/s}$$

Pump Work

$$W_{\text{act}} = p \times A$$

$$A = \frac{\pi}{4} D^2 = 0.07068 \text{ m}^2$$

$$p = 542857 \text{ N/m}^2$$

$$W.D_{\text{pump}} = p \times Q = 4071.35 \text{ Nm/s}$$

To find:

Total Work =?

Power Delivered to the appliance=?

□ **Total Work = $W.D_{\text{acc}} + W.D_{\text{pump}} = 5971.25 \text{ Nm/s}$**

□ **Power delivered = $\text{Total Work} / 1000 = 5.971 \text{ kW}$**

Solution:

8. An accumulator has a ram of diameter 250 mm and a lift of 8 m. The total weight on accumulator is 70 kN. The packing friction is 5% of the load on the ram. Find the power delivered to the machine if ram **falls through** the full height in 100 sec and at the same time the pumps are delivering 0.028 /s through the accumulator.

Given:

$$W = 70000 \text{ N}$$

$$D = 0.25 \text{ m}$$

$$\text{Distance moved (L)} = 8 \text{ m}$$

$$Q = 0.028 \text{ m}^3/\text{s}$$

$$\text{Frictional resistance} = 0.05$$

$$\text{Time} = 100 \text{ sec}$$

To find:

Power Delivered to the appliance=?

Solution:

Work of accumulator

$$\text{Actual weight} = W \times 0.95 = 66500 \text{ N}$$

$$W.D_{\text{acc}} = \text{-----} = 5320 \text{ Nm/s}$$

Pump Work

$$W_{\text{act}} = p \times A$$

$$A = \frac{\pi}{4} D^2 = 0.049 \text{ m}^2$$

$$p = 1354727 \text{ N/m}^2$$

$$W.D_{\text{pump}} = p \times Q = 37932 \text{ Nm/s}$$

$$\square \text{ Total Work} = W.D_{\text{acc}} + W.D_{\text{pump}} = 43252 \text{ Nm/s}$$

$$\square \text{ Power delivered} = \text{Total Work} / 1000 = 43.252 \text{ kW}$$

9. The diameters of fixed ram and fixed cylinder of an intensifier are 8 cm and 20 cm respectively. If the pressure of the water supplied to the fixed cylinder is 300 N/ , find the pressure of the water flowing through the fixed ram.

Given:

$$D = 0.2 \text{ m}$$

$$d = 0.08 \text{ m}$$

$$p = 300 \times 10^4 \text{ N/m}^2$$

To Find:

$$P^* = ?$$

Solution:

$$A = \frac{\pi}{4} D^2 = 0.0314 \text{ m}^2$$

$$a = \frac{\pi}{4} d^2 = 5.026 \times 10^{-3} \text{ m}^2$$

$$P^* = \frac{A}{a} p = 1875 \times 10^4 \text{ N/m}^2$$

10. The pressure intensity of water supplied to an intensifier is 20 N/ while the pressure intensity of water leaving the intensifier is 100 N/ The external diameter of the sliding cylinder is 20 cm. Find the diameter of the fixed ram of the intensifier.

Given:

$$D = 0.2 \text{ m}$$

$$p = 20 \times 10^4 \text{ N/m}^2$$

$$P^* = 100 \times 10^4 \text{ N/m}^2$$

To Find:

$$d = ?$$

Solution:

$$A = \frac{\pi}{4} D^2 = 0.0314 \text{ m}^2$$

$$P^* = \frac{A}{a} p$$

$$a = \frac{A}{P^*} p$$

$$d = 8.94 \text{ cm}$$

11. The water is supplied at the rate of 0.02m³/s from a height of 3 m to a hydraulic ram which raises 0.002m³/s to a height of 20 m from the ram. Determine D' Aubuisson's and Rankine's efficiencies of the hydraulic ram.

Given:

$$Q = 0.02 \text{ m}^3/\text{s}$$

$$h = 3 \text{ m}$$

$$q = 0.002 \text{ m}^3/\text{s}$$

$$H = 20 \text{ m}$$

To Find:

$$A_{\text{ub}} = ?$$

$$\text{Rank} = ?$$

Solution:

$$\square \text{ D' Aubuisson's efficiency}$$

$$A_{\text{ub}} = \frac{q}{Q} \times \frac{H}{h} = 66.67 \%$$

$$\square \text{ Rankine's efficiency}$$

$$\text{Rank} = \frac{q}{Q} \times \frac{H}{h} = 62.96 \%$$

12. The water is supplied at the rate of 3000 litres per minute from a height of 4m to a hydraulic ram which raises 300litres/minute to height of 30m from the ram. The length and diameter of the delivery pipe is 100m and 70mm respectively. Calculate the efficiency of the hydraulic ram if the co-efficient of friction $f=0.009$.

Given:

$$Q = 0.05 \text{ m}^3/\text{s}$$

$$h = 4 \text{ m}$$

$$q = 0.005 \text{ m}^3/\text{s}$$

$$H = 30 \text{ m}$$

$$L = 100 \text{ m}$$

$$d = 0.07 \text{ m}$$

$$f = 0.009$$

To Find:

$$A_{ub} = ?$$

$$Rank = ?$$

Solution:

Due to Effect of friction

$$\text{Effective head} = H + h_f$$

$$h_f = \text{---}$$

$$Q = a V$$

$$a = \frac{1}{4} d^2$$

$$V = 1.3 \text{ m/s}$$

$$h_f = 4.43 \text{ m}$$

$$\text{Effective head} = 34.43 \text{ m}$$

$$\square \text{ D' Aubuisson's efficiency}$$

$$A_{ub} = 86.07 \%$$

$$\square \text{ Rankine's efficiency}$$

$$Rank = 84.53 \%$$

13. A hydraulic lift is required to lift a load of 8 KN through a height of 10m, once in every 80 seconds. The speed of the lift is 0.5 m per second. Determine: (a) Power required to drive the lift (b) Working period of lift in seconds, (c) Idle period of the lift in seconds.

Given:

$$W = 8000 \text{ N}$$

$$\text{Distance moved (H)} = 10 \text{ m}$$

$$t = 80 \text{ s}$$

$$v = 0.5 \text{ m/s}$$

To Find:

$$\text{Power required} = ?$$

$$\text{Work period of lift} = ?$$

$$\text{Idle period of lift} = ?$$

Solution:

a. Power required to drive the lift:

$$W.D_{\text{lift}} = \text{---} = 1000 \text{ Nm/s}$$

$$\text{Power} = 1.0 \text{ kW}$$

b. Working period of the lift:

$$= 20 \text{ sec}$$

c. Idle period of the lift:

$$= \text{total time} - \text{working period of time}$$

$$= 60 \text{ sec}$$

14. A hydraulic lift is required to lift a load of 12 KN through a height of 10m, once in every 1.75 minutes. The speed of the lift is 0.75 m/s during working stroke of the lift, water from accumulator and the pump at a pressure of 400 N/ is supplied to the lift. If the efficiency of the pump is 8% and that of lift is 75%, find the power required to drive the pump and the minimum capacity of the accumulator. Neglect friction losses in the pipe.

Given:

$$W = 12000 \text{ N}$$

$$\text{Distance moved (H)} = 10 \text{ m}$$

$$t = 105 \text{ s}$$

$$v = 0.75 \text{ m/s}$$

$$p = 400 \times 10^4 \text{ N/m}^2$$

$$\eta_{\text{pump}} = 0.80$$

$$v_{\text{lift}} = 0.75$$

To Find:

$$\text{Power required to drive the pump} = ?$$

$$\text{Capacity of accumulator} = ?$$

Solution:

$$\text{Total Work done} = W.D_{\text{pump}} + W.D_{\text{acc}}$$

$$= \text{Total Weight} \times \text{distance moved per second}$$

$$= W \times v = 9000 \text{ Nm/s}$$

$$\text{Actual W.D} = \text{Total W.D} / \text{Lift} = 12000 \text{ Nm/s}$$

$$\text{W.D}_{\text{pump}} + \text{W.D}_{\text{acc}} = 12000 \text{-----} (1)$$

Working period of the lift

$$= \text{-----} = 13.33 \text{ sec}$$

Ideal period of the lift

$$= \text{Total time} - \text{working period} = 91.67 \text{ sec}$$

$$\text{W.D}_{\text{acc}} = \text{-----}$$

$$\text{W.D}_{\text{acc}} = 6.876 \text{ W.D}_{\text{pump}} \text{-----} (2)$$

Solving eqn (1) and (2)

$$\text{W.D}_{\text{pump}} = 1523.6$$

a. **Power required to drive the pump**

$$= \text{W.D}_{\text{pump}} / 1000 = 1.523 \text{ kW}$$

$$\text{Actual power required} = \text{power required} / \text{Pump}$$

$$= 1.90375 \text{ kW}$$

• **Minimum capacity of the accumulator**

$$= \text{W.D}_{\text{pump}} \times \text{Ideal time}$$

$$= 1523.6 \times 91.67 = 139613.4 \text{ Nm}$$

15. Find the efficiency of a hydraulic crane, which is supplied 300 litres of water under a pressure of 60 N/ for lifting a weight of 12 KN through a height of 11m.

Given:

$$V = 0.30 \text{ m}^3$$

$$p = 60 \times 10^4 \text{ N/m}^2$$

$$W = 12000 \text{ N}$$

$$\text{Distance moved (H)} = 11 \text{ m}$$

To Find:

$$\text{Efficiency} = ?$$

Solution:

$$\text{Output of the crane} = \text{Weight} \times \text{distance moved}$$

$$= 132000 \text{ Nm}$$

$$\text{Input of the crane} = p \times A \times L$$

$$V = A \times L$$

$$\text{Input of the crane} = 18 \times 10^4 \text{ Nm}$$

$$\text{Efficiency} = \text{output/input} = 73.33 \%$$

16. The efficiency of a hydraulic crane, which is supplied water under a pressure of 70 N/ for lifting a weight through a height of 10m is 60 %. If the diameter of the ram is 150mm and velocity ratio is 6, find,

(a) The weight lifted by the crane, (b) the volume of water required in litres to lift the weight.

Given:

$$\text{Efficiency} = 0.60$$

$$p = 70 \times 10^4 \text{ N/m}^2$$

$$D = 0.15 \text{ m}$$

$$\text{Distance moved (H)} = 10 \text{ m}$$

$$\text{Velocity ratio (H/L)} = 6$$

$$L = 1.667 \text{ m}$$

To Find:

$$W = ?$$

$$V = ?$$

Solution:

a. **Weight lifted by the crane:**

$$\text{Output of the crane} = \text{Weight} \times \text{distance moved}$$

$$= W \times 10$$

$$A = \frac{D^2}{4} = 0.01767 \text{ m}^2$$

$$\text{Input of the crane} = p \times A \times L$$

$$= 2.06 \times 10^4 \text{ Nm}$$

$$\text{Efficiency} = \text{output/input}$$

$$W = 1236.9 \text{ N}$$

b. **Volume of water:**

$$V = A \times L = 0.02945 \text{ m}^3$$

$$= 29.45 \text{ Litres}$$

17. In a hydraulic coupling the speed of the driving and driven shafts are 800 r.p.m and 780 r.p.m respectively. Find (a) the efficiency of the hydraulic coupling, (b) the slip of the coupling.

Given:

$$N_A = 800 \text{ r.p.m}$$

$$N_B = 780 \text{ r.p.m}$$

To Find:

$$\text{Efficiency} = ?$$

$$\text{Slip} = ?$$

Solution:

$$\text{Efficiency} = N_B / N_A$$

$$= 97.5 \%$$

$$\text{Slip} = (N_A - N_B) / N_A$$

$$= 0.025 = 2.5 \%$$

18. A hydraulic crane is lifting a weight of 12000 N through a height of 12m with a speed of 18m per minute once in every 2 minutes. The efficiency of the hydraulic crane is 65% and it is working under a pressure of 500 N/ of water. The crane is fed from an accumulator to which water is supplied by a pump. Find: (a) The capacity of the cylinder of the jigger in litres. (b) The capacity of the accumulator in litres (c) Minimum power required for the pump.

Given:

$$W = 12000 \text{ N}$$

$$\text{Efficiency} = 0.65$$

$$p = 500 \times 10^4 \text{ N/m}^2$$

$$\text{Distance moved (H)} = 12 \text{ m}$$

$$\text{Time} = 2 \text{ min}$$

$$v = 18 \text{ m/min}$$

To Find:

Volume of cylinder =?

Volume of accumulator =?

Minimum power required for the pump =?

Solution:

(a) **The capacity of the cylinder of the jigger in litres.**

$$\text{Output of the crane} = \text{Weight} \times \text{distance moved}$$

$$= 12000 \times 12$$

$$\text{Input of the crane} = p \times A \times L$$

$$\text{Efficiency} = \text{output/input}$$

$$\text{Volume} = A \times L = 0.0443 \text{ m}^3$$

$$= 44.3 \text{ Litres}$$

(b) **The capacity of the accumulator in litres**

$$\text{Volume of accumulator} = \text{volume of cylinder} \times -$$

$$= 29.53 \text{ litres}$$

(c) **Minimum power required for the pump.**

$$\text{Power} = \text{work input per minute} / 1000$$

$$\text{Work input per minute} = \text{—————}$$

$$= \text{—————}$$

$$= 1846 \text{ Nm/min}$$

$$\text{Power} = 1.856 \text{ kW}$$